Physics 4B Spring 2026
Week 8 Discussion

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1. **Brainwarmer/Adiabatic compression**[1]. What is the final T of an ideal gas initially at 20°C if compressed adiabatically to half its original volume? Assume $c_v = \frac{5}{2}R$.

   Solution: First we note that $c_v = \frac{5}{2}R$ means that this is a diatomic ideal gas, so if we ever need to compute the internal energy, it would have to be $U = \frac{5}{2}Nk_BT$ instead of $U = \frac{3}{2}Nk_BT$.

   We have an adiabatic process, so the first thing we should write down is $P_1V_1^\gamma = P_2V_2^\gamma$, with $\gamma = \frac{c_p}{c_v} = \frac{c_v+R}{c_v} = \frac{7}{5}$. We are given the initial temperature $T_1 = 293$ K and the final volume $V_2 = \frac{V_1}{2}$, so we can use the ideal gas law to replace the pressures with these given quantities:

   $$Nk_BT_1V_1^{\gamma-1} = Nk_BT_2V_2^{\gamma-1} \tag{1}$$

   Rearrange and replace V_2 with V_1, we get

   $$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1} = (293\text{ K})\,(2)^{2/5} = 387\text{ K} \tag{2}$$

   It makes sense that the temperature would rise when the gas is compressed without allowing heat to flow in or out.

2. **Isobaric and isochoric processes.**[2]. An ideal mono-atomic gas in state A ($P = 3\times10^5$ Pa, $V = 10^{-3}$ m^3, $U = 450$ J) is cooled at constant volume to B ($P = 2\times10^5$ Pa, $V = 10^{-3}$ m^3) and then expands at constant pressure to C ($P = 2\times10^5$ Pa, $V = 3\times10^{-3}$ m^3)[3]. What is

   (i) the work done,

   (ii) U_C,

   (iii) ΔQ[4], the heat added,

   (iv) the ratio of initial and final temperatures, T_C/T_A, and

[1] Shankar 11.22

[2] Shankar 11.17

[3] Nothing has been said about the path CA. So don't draw it.

[4] δQ in our notation.

(v) the ratio of internal energies U_C/U_A?

Solution:

(i) This problem consists of two processes: an isochoric ($\Delta V = 0$) process (A→B) and an isobaric ($\Delta P = 0$) expansion (B→C).

The work done *by* the gas is $\delta W = PdV$, so

$$W_{\mathrm{A}\to\mathrm{C}} = \int_A^C PdV = 0 + P_B(V_C - V_B) = 400 \text{ J} \tag{3}$$

(ii) The internal energy depends only on the temperature, which can be calculated from P and V using the ideal gas law, so

$$U_C = \frac{3}{2}Nk_BT_C = \frac{3}{2}Nk_B\frac{P_CV_C}{Nk_B} = \frac{3}{2}P_CV_C = 900 \text{ J} \tag{4}$$

(iii) Using the first law, we have

$$\begin{aligned}\delta Q_{\mathrm{A}\to\mathrm{C}} &= \Delta U_{\mathrm{A}\to\mathrm{C}} + \delta W_{\mathrm{A}\to\mathrm{C}} = (U_C - U_A) + W_{\mathrm{A}\to\mathrm{C}} \\ &= \frac{3}{2}P_CV_C - \frac{3}{2}P_AV_A + P_B(V_C - V_B) \\ &= 900 \text{ J} - 450 \text{ J} + 400 \text{ J} = 850 \text{ J}\end{aligned} \tag{5}$$

(iv) The ratio of initial and final temperatures is

$$\frac{T_C}{T_A} = \frac{P_CV_C}{P_AV_A} = 2 \tag{6}$$

(v) The ratio of internal energies is equal to the ratio of the temperatures, $\frac{U_C}{U_A} = \frac{T_C}{T_A} = 2$.

3. **Isothermal compression**[5]. A cylinder with a movable piston of cross-sectional area $A = 0.040$ m^2 contains an ideal gas initially at atmospheric pressure $P_1 = 1.013 \times 10^5$ Pa and temperature $T_1 = 273$ K. A 10 kg mass is slowly placed on the piston while the gas remains in thermal equilibrium with the surroundings.

(i) Find the percent decrease in the height of the gas column.

(ii) If the initial volume is $V_1 = 1$ m^3, find the work done by the gas and the heat flow into the gas during this compression.

(iii) The gas is then heated at constant pressure until the piston returns to its original height. Find the final temperature.

[5] Adapted from Shankar 11.21 and 11.23

Solution:

(i) The added mass provides a change in pressure

$$\Delta P = \frac{mg}{A} = \frac{(10\text{ kg})(9.8\text{ m/s}^2)}{0.040\text{ m}^2} = 2450\text{ Pa} \tag{7}$$

so $P_2 = P_1 + \Delta P = 1.038\ \times 10^5$ Pa.

For an isothermal process, $T_1 = T_2$, so the ideal gas law says $P_1V_1 = P_2V_2$. Also, $V = A \cdot h$, with A unchanged in the process, so

$$\frac{h_2}{h_1} = \frac{V_2}{V_1} = \frac{P_1}{P_2} = 0.976 \tag{8}$$

The percent decrease is therefore 2.4%.

(ii) The work done by the gas is

$$\begin{aligned} W &= \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \frac{Nk_BT_1}{V}dV = Nk_BT_1 \ln\left(\frac{V_2}{V_1}\right) \\ &= P_1V_1 \ln\left(\frac{P_1}{P_2}\right) = -2421\text{ J} \end{aligned} \tag{9}$$

The gas did negative work because it's compressed.

The heat flow into the gas is

$$\delta Q = \Delta U + \delta W = 0 + (-2421\text{ J}) = -2421\text{ J} \tag{10}$$

When compressed, the gas should heat up if the process is adiabatic, but heat flows out to keep the temperature constant.

(iii) We have an isobaric process, with $P_2 = P_3$. We are also given that $V_3 = V_1$, so

$$\frac{V_3}{V_2} = \frac{T_3}{T_2} = \frac{V_1}{V_2} = \frac{T_3}{T_1} \tag{11}$$

The final temperature is therefore

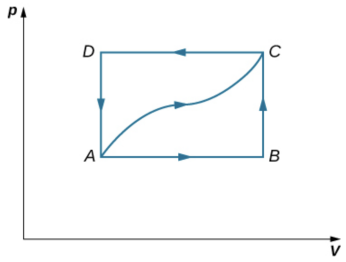
$$T_3 = \frac{V_1}{V_2}T_1 = \frac{273\text{ K}}{0.976} = 280\text{ K} \tag{12}$$

4. **Path dependence of work and heat**[6]. When a gas expands along AB (see below), it does 20 J of work and absorbs 30 J of heat. When the gas expands along AC, it does 40 J of work and absorbs 70 J of heat.

(a) How much heat does the gas exchange along BC?

[6]University Physics Volume 2 Chapter 3 Problem 45

(b) When the gas makes the transition from C to A along CDA, 60 J of work are done on it from C to D. How much heat does it exchange along CDA?



Solution:

(a) The change in internal energy from A to C is the same regardless of the path taken. Using the first law, we have

$$\Delta U_{\mathrm{A}\to\mathrm{C}} = \delta Q_{\mathrm{AC}} - \delta W_{\mathrm{AC}} = 70\ \mathrm{J} - 40\ \mathrm{J} = 30\ \mathrm{J} \tag{13}$$

Similarly, we have

$$\Delta U_{\mathrm{A}\to\mathrm{B}} = \delta Q_{\mathrm{AB}} - \delta W_{\mathrm{AB}} = 30\ \mathrm{J} - 20\ \mathrm{J} = 10\ \mathrm{J} \tag{14}$$

The change in internal energy from B to C is the difference between the two: $\Delta U_{\mathrm{B}\to\mathrm{C}} = \Delta U_{\mathrm{A}\to\mathrm{C}} - \Delta U_{\mathrm{A}\to\mathrm{B}} = 20$ J. There's no work done by the gas from B to C since this is an isobaric process, so the heat that flows into the gas from B to C is equal to the change in internal energy:

$$\delta Q_{\mathrm{BC}} = \Delta U_{\mathrm{B}\to\mathrm{C}} = 20\ \mathrm{J} \tag{15}$$

(b) We already calculated the change in internal energy from C to A, which is $\Delta U_{\mathrm{C}\to\mathrm{A}} = -\Delta U_{\mathrm{A}\to\mathrm{C}} = -30$ J. No work is done by the gas from D to A since that's an isobaric process, so the total work done *by* the gas from C to D is -60 J. Therefore, the first law says the heat that goes in the gas along CDA is

$$\delta Q_{\mathrm{CDA}} = \Delta U_{\mathrm{C}\to\mathrm{A}} + \delta W_{\mathrm{CDA}} = -30\ \mathrm{J} + (-60\ \mathrm{J}) = -90\ \mathrm{J} \tag{16}$$

This means 90 J of heat is released by the gas as it goes from C to A along CDA.